

View Review

Paper ID

801

Paper Title

AI-Driven Ensemble for Enhanced Sentiment Polarity Detection in Movie Reviews

REVIEW QUESTIONS

1. Comments to Authors

A modular ensemble framework of sentiment polarity detection with generative, discriminative, and deep learning modeling is presented in this paper. The strategy has strong motivation, and it consistently improves the performance on commonly used benchmarks. The weighted voting mechanism and fallback abstention strategy are considered prudent addition, which improves robustness and interpretability.

Nonetheless, there are several crucial facets that need to be addressed prior to publication:

- ☐ Although baseline comparisons are included, there is a lack of statistical significance testing (e.g., paired t-tests, confidence intervals). This restricts the force of assertions regarding performance.
- ☐ The internal dynamics of the ensemble are not examined. By performing a component-wise ablation, it would become clearer how each model contributes and the fusion strategy would be validated.
- ☐ SHAP and LIME are referenced but not put into practice. Interpretability at the token level or based on attention would bolster trust and transparency.
- ☐ There is no evidence to support claims of cross-domain robustness. Testing on other datasets (such as tweets and product reviews) would confirm generalizability.
- ☐ Adaptive or trainable fusion methods (like stacking, gating networks) could improve the static weighted voting system.

2. Recommendation: Your overall rating

- Major Revision
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